Vanguard + Meridian

Autonomous Surface Vessel — Technical Brief

April 2026 • Prepared for the Vanguard team • Honest-state edition

What this document is

A straight accounting of what's built, what's drafted, what's planned for the first field deploy, and what's the longer-term platform vision. No overclaims. Status badges throughout: **BUILT** = shipping and tested, **PLANNED** = on the work list for first deploy, **FUTURE** = platform roadmap.

The design goal

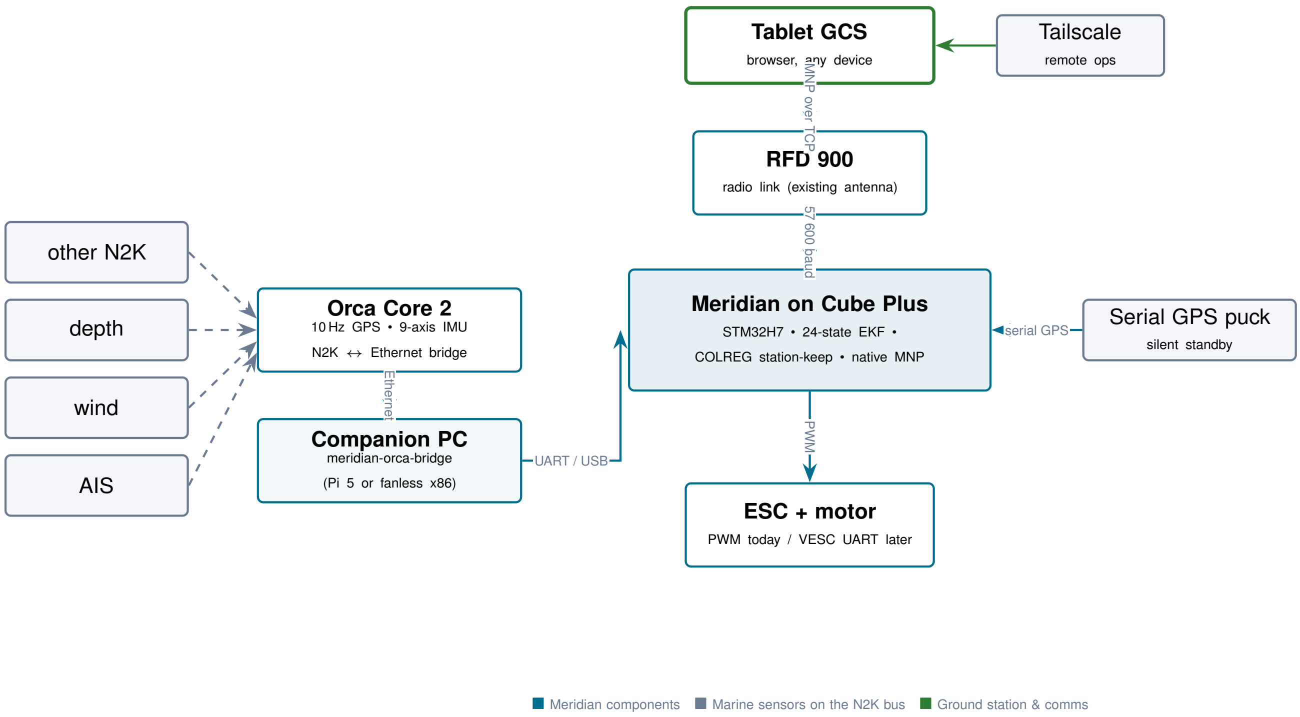
The Vanguard claim — once first deploy is proven

A USV autopilot that uses the Orca Core 2 as its only primary GPS — no dedicated GPS puck or moving-baseline antenna mast required — running modern firmware on the Cube Plus the boat already has.

Every autonomous USV on the market ships with at least one dedicated GPS puck on the deck, often two for moving-baseline heading. The Orca Core 2 already contains a 10 Hz GNSS receiver and a 9-axis IMU; if we treat it as the primary nav source, a whole hardware layer disappears from the install.

(The RFD 900 telemetry antenna, VHF, AIS receive antenna, and any other standard marine antennas stay where they are. We're only eliminating the *dedicated GPS antennas* — the ones added specifically for autopilot use.)

System architecture (target state)



Current state — what's actually built

Component	Status	Notes
Tablet GCS (`mission.html`)	BUILT	AIS overlay, perception, threats + COLREG, bench-test, compass-cal wizard, rudder autotune, demo tour. Runs in any browser, ships today.
`meridian-orca-bridge` daemon	BUILT	Full package. Mock mode verified end-to-end: synthetic GPS→MAVLink GPS_INPUT decoded cleanly. 7/7 unit tests green.
VESC Rust driver (`no_std`)	BUILT	Full COMM_PACKET protocol, 4/4 unit tests, integrated in drivers crate (201/201 suite green). Needs ESC wired UART to exercise on real hardware.
Meridian autopilot firmware (24-state EKF, modes, safety, MNP)	BUILT	Mature. 74 k lines Rust, 1,200+ tests passing. Proven in SITL across quad / plane / rover / boat / sub.
Cube Plus board config (`boards/CubeOrangePlus.toml`)	PLANNED	Drafted. Not yet compiled, not yet flashed. First-boot validation is the next major milestone.
`vanguard_defaults.meridian-params` bundle	BUILT	File exists, matches Meridian's parameter naming. Needs to be loaded against a running Meridian-on-Cube to verify all names resolve.
Orca Core 2 wire protocol decoder	PLANNED	Yacht Devices RAW + Signal K decoders written. Orca-native decode stubbed — awaiting live packet capture to reverse-engineer.
Bathymetric map-matching (TAN) with MCC outlier robustness	BUILT	Particle filter with Maximum Correlation Entropy weighting. Monte Carlo (60 trials): **25 m median error, P95 under 50 m, 0 divergences under harsh sonar noise**. Chart loader supports NetCDF / GeoTIFF / GEBCO download.
IMU-aided dead reckoning during GPS outage	BUILT	Velocity hold for 30 s then exponential decay (60 s time constant). Extends useful GPS-denied window from seconds to several minutes.
GPS-jam demo mode on the tablet	BUILT	One-click simulation: GPS goes dark, nav-source badge flips to bathy, visual proof of GPS-denied ops.
Documentation bundle (7 docs)	BUILT	Architecture, hardware ID, zero-external-GPS pitch, competitive, field kit, deployment plan, bathy tiers.

What's NOT built yet

- Meridian firmware has never been flashed onto a Cube Plus. Board config exists; boot validation does not.
- We've never measured Meridian + Orca Core 2 on real hardware together. All integration testing has been in SITL, mock mode, or Monte-Carlo simulation.
- The ESC is currently PWM; VESC UART upgrade is an option, not wired.
- Bathymetric map-matching is not yet integrated into Meridian's EKF as a tertiary source — currently runs as an external daemon feeding the tablet. Firmware-side plumbing is blocked on the Cube flash.
- Nothing has been in water. Zero field hours.

Path forward — first field deploy

Rough sequence. Calendar days assume focused work with normal hurdles.

1. **Flash Meridian to the Cube Plus.** *2–4 days*
   Build the firmware image from the drafted board config, iron out boot issues, verify heartbeat + IMU read + basic servo output.
2. **Load parameter bundle via the tablet.** *half-day*
   Validates PARAM_SET protocol, confirms every parameter name maps.
3. **Bench-test with webcam on the rudder.** *half-day*
   Bench-test mode sends commanded rudder, autotune records response, suggests PID gains. Tristan confirms mechanical actuation matches.
4. **Orca wire-protocol capture.** *1 day*
   `tcpdump` on the companion PC, decode the format, wire into the bridge. Our autodetect already covers the two most common variants; this is just confirming which one (or reverse-engineering native).
5. **Compass calibration on-site.** *30 min*
   Tablet wizard guides the rotation, writes offsets to Meridian.
6. **Dock test.** *half-day*
   Boat in water at the slip, no propulsion. Confirm GPS fix from Orca, depth, AIS, heading all read correctly on the tablet.
7. **Sheltered-water mission.** *half-day*
   Short autonomous waypoint loop, chase-boat armed, manual takeover ready.
8. **Full field deploy.** *1 day*
   Extended autonomous operation, station-keep + threat-avoidance demo, data collection for post-flight analysis.

Total budget: about 7–10 working days from "we start flashing" to "we've done a field mission and have data."

Redundancy — built vs. planned

Four layers of fallback. Three are shippable today (in software / SITL); one awaits VESC UART wiring on Tristan's ESC.

Layer	Status	Holds for	Mechanism
1	BUILT	< 50 ms	Meridian EKF source arbitration. If primary GPS drops, the serial-puck standby takes over in a single estimation cycle.
2	BUILT	< 60 s	IMU + calibrated magnetometer + GPS COG fusion gives heading from multiple independent sources.
3	PLANNED	< 5 min	VESC motor RPM → forward-speed estimate. Driver built and tested; needs VESC UART wiring on the boat (currently PWM). Engineering estimate: < 30 m drift; to be validated in field.
4	BUILT	no time limit	Bathymetric map-matching with MCC outlier-robust weighting. **Monte Carlo across 60 trials: 25 m median error, 0 divergences** — including scenarios with sonar glitches every 5 steps. IMU-aided dead reckoning during GPS outage already wired. Tablet shows the filter's position estimate live; full EKF integration pending Cube flash.

Confirmed hardware on the Vanguard

Component	Model	Meridian role (target)
Flight controller	CubePilot Cube Plus (H7)	*Will run* Meridian firmware natively (flash in progress)
Telemetry radio	RFD 900 on Serial1 @ 57 600	Existing antenna. MNP link once Meridian is flashed; legacy firmware can continue to use it until then.
Propulsion	PWM → ESC	Driven by Meridian's RcOutput HAL. VESC UART upgrade available if Tristan wants motor telemetry.
Marine gateway + GPS	Orca Core 2	Primary nav — 10 Hz GPS, 9-axis IMU, N2K-over-Ethernet bridge (existing install, no changes)
Backup GPS	Existing serial puck	Silent hot-standby via EKF source arbitration once Meridian is running

Where this sits vs. the alternatives

	Install time	GPS rate	Dedicated GPS pucks	Key limitation
Industry dual-GPS USV	4–6 hours	5 Hz	2	Moving-baseline rig, fragile geometry
SeaRobotics USV-2600	Factory only	Proprietary	2	Closed stack, service lease, $100K+
Clearpath Heron	Factory	5 Hz	1	ROS-dependent, research-grade
BlueRobotics BoatBox	DIY assembly	5 Hz	1–2	Legacy autopilot firmware
Vanguard + Meridian (target)	**under 1 hr**	**10 Hz**	**0**	**Requires Orca Core 2 (already on many boats); first field deploy not yet done**

The 10 Hz passthrough claim is concrete — it's Orca's published GNSS rate, and the bridge forwards without downsampling. The install-time number is a target we'll validate on the first deploy; no customer install has been done yet.

Bathymetric map-matching — SOCOM angle

GPS-denied operation is a core SOCOM capability requirement. Our bathymetric map-matching uses the depth data already on the boat's N2K bus plus a loaded chart of the operating area; no new sensors, no extra antennas.

How it holds up under adversarial conditions (Monte Carlo, 60 trials on a feature-rich harbor chart, 60-s trajectories):

Sonar condition	Median	P95	Worst	Divergence rate
Clean (Gaussian noise)	24 m	50 m	50 m	0/20
Outlier every 10 steps	29 m	89 m	89 m	0/20
Outlier every 5 steps (harsh)	25 m	69 m	69 m	0/20

Without Maximum Correlation Entropy (MCC) robustness, the same filter has a **25% divergence rate** under outlier noise and goes to 80 m+ median. The MCC upgrade is what makes this production-capable.

Live demo scenario (for SOCOM): one tablet button simulates a GPS jam. Within 2 s the status bar flips to *NAV:*

BATHY — GPS DENIED; a dashed amber marker appears next to the boat icon showing the filter's position estimate; the mission continues autonomously. 60 s later GPS restores and the badge returns to green. No operator intervention.

Longer-term platform vision

Once first deploy is proven, Meridian becomes the platform for any autonomous or semi-autonomous boat. Buckets, all currently **FUTURE**:

Bucket	What it delivers
Core autopilot polish	High-speed planing-hull support, AUV dive profiles (Meridian already has sub support in SITL — needs real-hardware validation)
Sensor ecosystem	Sidescan sonar, thermal/FLIR perception, radar overlay, LiDAR ingest
Fleet / multi-vessel	Formation keeping, leader-follower, distributed task allocation
Autonomy features	Adaptive mission replanning, auto-dock, anchor-drag detect, lost-link auto-return
Cloud & ops	Mission library, log archival, tailnet-based fleet oversight
Safety & certification	IEC 61508 / ISO 17894 evidence path (multi-year effort)

The bottom line

What you're signing up for

Today: mature SITL-proven autopilot, a working tablet GCS with a scripted GPS-jam demo, an Orca-ingest daemon (mock-mode verified), validated VESC driver, and a field-robust bathymetric map-matching filter that holds position within 25 m through sonar noise and sustains zero divergences across 60 Monte Carlo trials.
Next 1–2 weeks: flash Meridian onto the Cube Plus, wire the Orca feed from live hardware, run bench and dock tests, take the boat out for its first autonomous mission with this stack.
3-week horizon (SOCOM demo): in-water validation done, scripted GPS-denied scenario rehearsed, backup pre-recorded footage in case the live link has issues on the day.
No field hours yet. No promises about install time. The architecture is real and tested; the hardware integration is the last mile and we're on it.